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Parameter identification of Sub-synchronous/Super-synchronous oscillations based on synchrophasor rotation and spectral shift

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ABSTRACT

Benefiting from the advantage of a unified time scale of measurement data from different sites, the sub-synchronous oscillation (sub-SO) and super-synchronous oscillation (super-SO) detection and mode parameter identification based on a wide-area measurement system can be applied to analyze the development process of oscillation. The phasor measurement units report rate constrains the existing methods, indicating that they can only identify the sub-SO parameters and cannot be extended to super-SO. This study proposes a novel method for sub-SO and super-SO detection and parameter identification based on synchrophasor rotation characteristics and spectral shift. Analyzing the spectral superposition of sub-SO and super-SO components in the real part of the synchrophasor is crucial for realizing sub-SO and super-SO detection. Subsequently, the fundamental, sub-SO, and super-SO components in the synchrophasor are decoupled using the rotational characteristics of the synchrophasor to identify parameters accurately. Compared with the existing methods, the proposed method can realize the expansion of the parameter identification range to super-SO. The effectiveness of the proposed method is verified by simulation and practical engineering cases.

1. Introduction

In recent years, the number of power electronic devices connected to the power grid has increased with the continuous increase of renewable energy capacity. The interaction of power electronic devices and the resonant circuit of the power system (such as series compensation) causes sub-synchronous oscillation (sub-SO) and super-synchronous oscillation (super-SO), which seriously endanger the security and stability of a power system [1,2]. In addition, oscillation energy propagates from local to global in the power grid, showing the characteristics of wide-area propagation [3,4], which is not conducive to accident analysis and oscillation suppression [5,6]. Therefore, identifying sub-SO and super-SO mode parameters efficiently is vital for accident recurrence, traceability [7,8], and real-time suppression [9] because the mode parameter can reflect the key characteristics of sub-SO and super-SO.

1.1. Review of previous publications

Various methods have been proposed to identify oscillation mode parameters, which can be separated into two categories. One category mainly includes the discrete Fourier transform (DFT)-based method [10,11], wavelet transform method [12], Prony method [13], subspace class algorithms [14], and mode decomposition method [15]. They all use instantaneous data recorded from phasor measurement units (PMUs) with sampling frequencies of up to 10 kHz. These methods have the advantages of low complexity, wide measurement frequency band, and high measurement accuracy. However, these methods, which rely on the instantaneous data recorded by PMUs, can only be applied for postevent analysis. Moreover, they fail to perform real-time analysis of the propagation and development of oscillation at the system level.

The other category uses synchrophasor data from the wide-area measurement system (WAMS) main station. The WAMS main station collects synchrophasor data reported by crucial nodes to realize wide-area monitoring and real-time analysis of sub-SO and super-SO [8,16,17,18,19], and [20]. In [16], a table look-up algorithm based on synchrophasor amplitude spectrum analysis was proposed. This method makes the spectral leakage amplitude of the sub-SO component in the synchrophasor correspond to the sub-SO amplitude. It also realizes the accurate identification of the amplitude and frequency. [17] and [18] adopted Hann-windowed DFT and interpolated DFT to process

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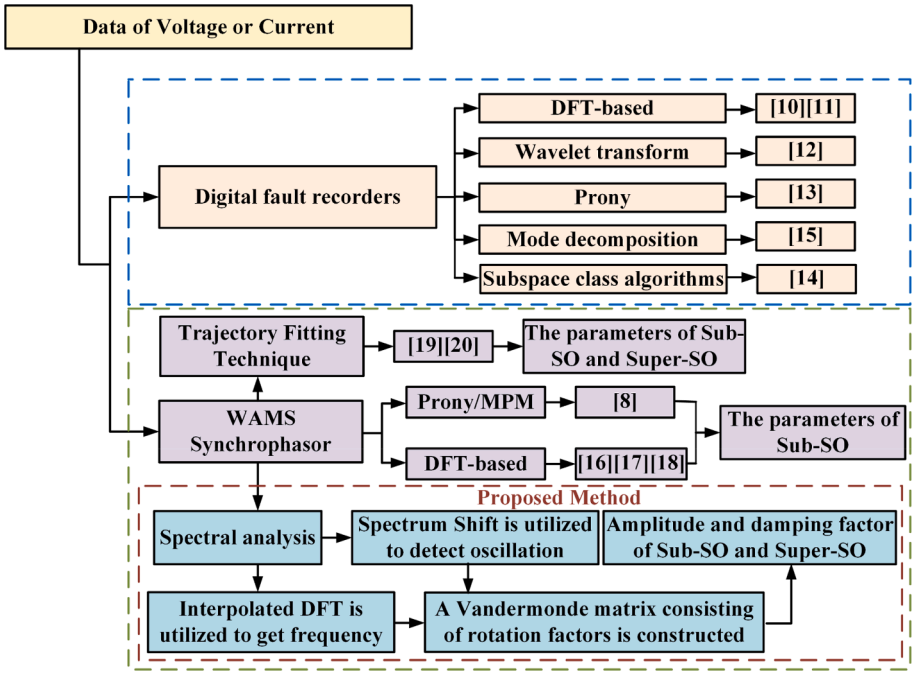


Fig. 1. Comparisons between previous publications and the proposed method.

synchrophasor, overcoming the problem of low spectral resolution caused by the PMU report rate. The damping factor parameter was also obtained. In addition, the classical methods of Prony and the matrix pencil were simultaneously introduced in [8], realizing the transformation from sub-SO mode parameter identification to a mode parameter extraction. Sub-SO interference source localization was also realized. However, the above methods are all limited by the PMU report rate. Moreover, these methods only identify the mode parameters of the sub-SO. The identification error increases significantly once the frequency-coupled super-SO exists in the system.

A synchrophasor trajectory fitting method was further proposed [19] to address this issue by realizing the parameter identification of sub-SO and super-SO amplitude and frequency. Furthermore, [20] proposed an improved method of interpolated DFT to solve the problem in [19], in which the damping factor could not be determined. However, these methods require a preset signal model and fail to handle the presence of multiple coupled sub-SO and super-SO components in the signal.

1.2. Description of the proposed method

In the present work, the proposed method realizes sub-SO and super-SO detection and mode parameter identification by utilizing synchrophasor rotational characteristics and spectral shifts. The spectral shift characteristic refers to the spectrum of the component in the signal that is shifted and superimposed with the spectrum of another component. Spectral shift essence is a kind of spectral mixing that occurs because the Nyquist sampling theorem is not satisfied. In Fig. 1, the proposed method is contrasted with previously published methods from the literature.

1.3. Summary of contributions

The summary of contributions of the present study is as follows.

(1) A Vandermonde matrix is created out of rotation factors to decouple and separate the fundamental, sub-SO, and super-SO components of the synchrophasor. In addition, the mode parameter identification of sub-SO and super-SO is accomplished by an iterative process. The proposed method can be extended to the systems with uncoupled sub-SO and super-SO components or multiple pairs of coupled sub-SO and super-SO components.
(2) The spectral shift characteristics of the super-SO component in the real part of the synchrophasor are analyzed to realize sub-SO and super-SO detection, revealing the spectral superposition relationship between the super-SO component and sub-SO component when the reporting rate f_r is twice the fundamental frequency f_N (50 or 60 Hz).
(3) The relationship between the oscillation mode parameters and the mode parameters of each component in the real part of the synchrophasor is obtained. The interpolated DFT is introduced to analyze the spectrum of the real part of the synchrophasor. The sub-SO and super-SO frequencies can be accurately identified.

This paper is organized as follows. The correspondence between synchrophasor and the existence of time-domain oscillatory signals is derived in Section 2. Section 3 presents a detection and mode parameter identification method for sub-SO and super-SO. The verification with both the synthetic and simulated data is illustrated in Section 4. Section 5 verifies the effectiveness of the proposed method through an actual engineering case. Finally, the conclusions are drawn in Section 6.

2. Representation of the sub-SO/super-SO components in synchrophasor

This section analyzes the sub-SO and super-SO components in the synchrophasor. The voltage and current signals $x(t)$ during sub-SO and super-SO of the power system are generally modeled as

$$x(t) = \sum_{i=0}^{2} A_i e^{\alpha_i t} \cos(2\pi f_i t + \varphi_i) \tag{1}$$

where A_i, f_i, α_i, and φ_i are amplitudes, frequencies, damping factors, and initial phases of the ith mode, respectively. $x(t)$ consists of three modes, including the fundamental component, the sub-SO component, and the super-SO component. The sub-SO component frequency f_1 is coupled with the super-SO component frequency f_2, expressed as $f_1 + f_2 = 2f_N$.

The PMUs sample the signal at a fixed frequency f_p and discretize the signal $x(t)$. Then, the nth sample data can be expressed as

$$x(n) = \sum_{i=0}^{2} A_i e^{\alpha_i \frac{n}{f_p}} \cos(2\pi f_i \frac{n}{f_p} + \varphi_i) \tag{2}$$

where f_p is the sampling frequency. $x(n)$ is truncated by a rectangular window of one fundamental period. Then, the synchrophasor is obtained by applying the DFT. The DFT in the rth sliding window can be expressed as

$$X(r,k) = \frac{1}{N_p}\sum_{i=0}^{2} A_i e^{r\frac{\alpha_i}{f_p}} \sum_{n=0}^{N_p-1} (e^{n\frac{\alpha_i}{f_p}} \cos(2\pi f_i \frac{(r+n)}{f_p} + \varphi_i) e^{-j2\pi\frac{kn}{N_p}} \quad r = 1,\ 2,\ \ldots,\ \ k = 0,\ 1,\ 2,\ \ldots,\ \ N_p\text{-}1, \tag{3}$$

where $N_p = f_p / f_N$ is the length of the rectangular window, and k is the number of spectral bins in Eq. (3). The expression can be rewritten as Eq. (4) by Euler equation and series summation methods.

$$X(r,k) = \frac{1}{2N_p}\sum_{i=0}^{2} A_i e^{r\frac{\alpha_i}{f_p}} (e^{j(2\pi r\frac{f_i}{f_p}+\varphi_i)} H(\alpha_i, \frac{f_i}{f_N} - k) + e^{-j(2\pi r\frac{f_i}{f_p}+\varphi_i)} H^*(\alpha_i, \frac{f_i}{f_N} + k)) \tag{4}$$

with

$$H(\alpha_i, \frac{f_i}{f_N} - k) = \frac{1 - e^{(\frac{\alpha_i}{f_N} + j2\pi(\frac{f_i}{f_N} - k))}}{1 - e^{(\frac{\alpha_i}{f_N} + j2\pi(\frac{f_i}{f_N} - k))/N_p}} \tag{5}$$

where $(\cdot)^*$ denotes the conjugate operator. The spectral bin corresponding to the fundamental frequency ($k = 1$) is reported as synchrophasor, expressed as

$$X(r) = X(r,1) = \frac{1}{2N_p}\sum_{i=0}^{2} A_i e^{r\frac{\alpha_i}{f_p}} (e^{j(2\pi r\frac{f_i}{f_p}+\varphi_i)} H(\alpha_i, \frac{f_i}{f_N} - 1) + e^{-j(2\pi r\frac{f_i}{f_p}+\varphi_i)} H^*(\alpha_i, \frac{f_i}{f_N} + 1)) . \tag{6}$$

The synchrophasor $X(r)$ contains the spectral leakage information of the components of the fundamental frequency, sub-SO, and super-SO, which are linearly superimposed. PMUs convert the synchrophasor into amplitude and phase angle form and report it to the WAMS main station with the timestamp, where the amplitude is expressed as

$$|X(r)| = \sqrt{\left|\sum_{i=0}^{2} Re(X(r))\right|^2 + \left|\sum_{i=0}^{2} Im(X(r))\right|^2} . \tag{7}$$

The phase angle λ is expressed as

$$\lambda = \arctan(\frac{\sum_{i=0}^{2} Im(X(r))}{\sum_{i=0}^{2} Re(X(r))}) . \tag{8}$$

This work aims to detect sub-SO and super-SO from synchrophasor and identify their mode parameters. As shown in Eqs. (6), (7), and (8), the synchrophasor amplitudes obtained by the WAMS main station contain information about different modes. The spectrum analysis of synchrophasor amplitude shows that the spectrum contains the f_i (i = 0,1,2) component. It also contains other frequency components, such as $f_i \pm f_j$ (i = 0,1,2, j = 0,1,2). Therefore, the sub-SO and super-SO mode parameter identification is difficult to realize based on the synchrophasor amplitude information.

α_i and f_i are typically considered constant over multiple consecutive short time windows. Therefore, $H(\alpha_i, f_i/f_N - 1) = H_{i1}e^{j\theta_{i1}}$ and $H^*(\alpha_i, f_i/f_N + 1) = H_{i2}e^{-j\theta_2}$ are set, and the Euler equation and trigonometric formula are utilized to rewrite Eq. (6) as

$$X(r) = \frac{1}{2N_p}\sum_{i=0}^{2} A_i e^{r\frac{\alpha_i}{f_p}} \left(e^{j(2\pi r\frac{f_i}{f_p}+\varphi_i)} H_{i1} e^{j\theta_{i1}} + e^{-j(2\pi r\frac{f_i}{f_p}+\varphi_i)} H_{i2} e^{-j\theta_{i2}}\right)$$
$$= \frac{1}{2N_p}\sum_{i=0}^{2} A_i e^{\frac{\alpha_i}{f_p}r} (F_{i1}\cos(2\pi r\frac{f_i}{f_p} + \varphi_i + \phi_{i1} + jF_{i2}\sin(2\pi r\frac{f_i}{f_p} + \varphi_i + \phi_{i2})) \tag{9}$$

with

$$F_{i1} = \sqrt{H_{i1}^2 + H_{i2}^2 + 2H_{i1}H_{i2}\cos(\theta_{i1} - \theta_{i2})} \tag{10}$$

and

$$\phi_{i1} = \arctan(\frac{H_{i1}\sin(\theta_{i1}) + H_{i2}\sin(\theta_{i2})}{H_{i1}\cos(\theta_{i1}) + H_{i2}\cos(\theta_{i2})}) . \tag{11}$$

According to Eqs. (9) – (11), the relationship between time-domain sub-SO and super-SO and the real and imaginary parts of the synchrophasor is indicated. Sub-SO and super-SO appear in the synchrophasor in the form of spectral leakage, which is the basis for implementing the proposed method. Analyzing the super-SO component is challenging because of the limitation of the Nyquist sampling theorem and the report rate of synchrophasor. However, some pieces of information about the super-SO component remain in the synchrophasor. Therefore, synchrophasor-based sub-SO and super-SO detection and parameter identification can be realized.

3. Sub-SO/Super-SO detection and mode parameter identification method

According to Section 2, the synchrophasor contains information about sub-SO and super-SO. Therefore, the spectral shift is used in this section to detect sub-SO and super-SO. The synchrophasor rotation characteristic is used to decouple and separate the fundamental, sub-SO, and super-SO components in the synchrophasor.

3.1. Spectral shift characteristics of super-SO

PMUs report synchrophasor to the WAMS main station at a reporting rate f_r below $2f_N$. Then, the synchrophasor data $X_n(m)$ of the WAMS main station is obtained by sampling $X(r)$ at the reporting rate f_r, which is expressed as

$$X_n(m) = X(mf_p/f_r) . \tag{12}$$

Then, Eq. (12) is substituted into Eq. (10), thereby obtaining

$$X_n(m) = \frac{1}{2N_p}\sum_{i=0}^{2} A_i e^{m\frac{\alpha_i}{f_r}} (e^{j(2\pi m\frac{f_i}{f_r}+\varphi_i)} H_{i1} e^{j\theta_{i1}} + e^{-j(2\pi m\frac{f_i}{f_r}+\varphi_i)} H_{i2} e^{-j\theta_{i2}})$$
$$= \frac{1}{2N_p}\sum_{i=0}^{2} A_i e^{m\frac{\alpha_i}{f_r}} (F_{i1}\cos(2\pi m\frac{f_i}{f_r} + \varphi_i + \phi_{i1}) + jF_{i2}\sin(2\pi m\frac{f_i}{f_r} + \varphi_i + \phi_{i2})) . \tag{13}$$

According to Eq. (14), the real part of the synchrophasor is calculated by utilizing the amplitude and phase information of the WAMS main station, which is expressed as

$$Re(X_n(m)) = |X_n(m)|\cos(\lambda) = \frac{1}{2N_p}\sum_{i=0}^{2} A_i F_{i1} e^{\alpha_i m/f_r} (\cos(2\pi m f_i/f_r + \varphi_i + \phi_{i1})) . \tag{14}$$

The proposed method aims to detect the existence of sub-SO and super-SO in the power system by performing spectrum analysis on the real part of the synchrophasor.

When $f_r = 2f_N$, the sub-SO and super-SO component frequency coupling characteristic $f_1 + f_2 = 2f_N$ and trigonometric formula are used; Eq. (14) can be rewritten as

$$Re(X_n(m)) = \frac{1}{2N_p}(A_0F_{01}e^{m\frac{\alpha_0}{f_r}}\cos(2\pi m\frac{f_0}{f_r} + \varphi_0 + \phi_{01}) + A_1F_{11}e^{m\frac{\alpha_1}{f_r}}(\cos(2\pi m\frac{f_1}{f_r} + \varphi_1 + \phi_{11}) + A_2F_{21}e^{m\frac{\alpha_2}{f_r}}(\cos(2\pi m\frac{f_1}{f_r} + \varphi_2 + \phi_{21})) . \tag{15}$$

According to Eq. (15), the super-SO component occurs a spectral

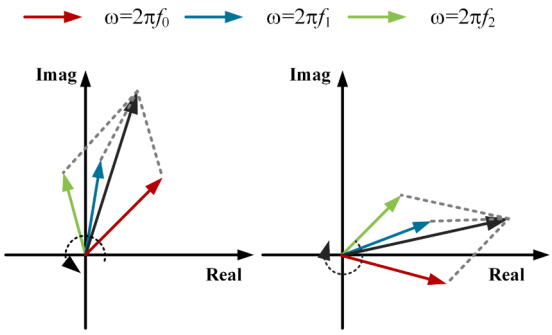


Fig. 2. Rotation characteristics of synchrophasor.

shift, which is superposed with the sub-SO component spectral. Therefore, when the spectrum analysis of the real part of the synchrophasor is performed and the components appear in the spectrum, three kinds of time-domain oscillations may appear:

1) Only sub-SO exists in the power system;
2) Only super-SO exists in the power system;
3) Frequency-coupled sub-SO and super-SO exist in the power system.

When $f_r = 100$ Hz, the frequency resolution of the spectrum analysis is 1 Hz, and the frequency results obtained by the spectral analysis may have serious estimation errors. Therefore, the interpolated DFT is used to improve the frequency identification accuracy of sub-SO and super-SO [18]. According to the frequency-coupling characteristics of sub-SO and super-SO, the super-SO frequency can be obtained.

3.2. Sub-SO/super-SO component decoupling and mode parameter identification

In Eq. (13), the component with frequency f_i in synchrophasor consists of two parts, which rotates with a positive angular frequency $2\pi f_i$ and a negative angular frequency $-2\pi f_i$. The rotation characteristics are shown in Fig. 2.

Given the rotation characteristics, Eq. (13) can be expressed as

$$\mathbf{B}\begin{bmatrix} A_0H_{01}e^{j(\varphi_0+\theta_{01})} \\ A_0H_{02}e^{-j(\varphi_0+\theta_{02})} \\ A_1H_{11}e^{j(\varphi_1+\theta_{11})} \\ A_1H_{12}e^{-j(\varphi_1+\theta_{12})} \\ A_2H_{21}e^{j(\varphi_2+\theta_{21})} \\ A_2H_{22}e^{-j(\varphi_2+\theta_{22})} \end{bmatrix} = \begin{bmatrix} X_n(m) \\ X_n(m+1) \\ X_n(m+2) \\ X_n(m+3) \\ X_n(m+4) \\ X_n(m+5) \end{bmatrix} \tag{16}$$

where **B** is the Vandermonde matrix constructed from rotation factors $e^{j2\pi f/f_r}$, and $X_n(m)$ is the synchrophasor. Therefore, $X_n(m)$ is a known quantity in Eq. (16). **B** represents the positive and negative rotation characteristics of each frequency component in the synchrophasor. **B** can be expressed as

$$\mathbf{B} = \frac{1}{2N_p}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ e^{(\alpha_0+j2\pi f_0)/f_r} & e^{(\alpha_0-j2\pi f_0)/f_r} & e^{(\alpha_1+j2\pi f_1)/f_r} & e^{(\alpha_1-j2\pi f_1)/f_r} & e^{(\alpha_2+j2\pi f_2)/f_r} & e^{(\alpha_2-j2\pi f_2)/f_r} \\ e^{2(\alpha_0+j2\pi f_0)/f_r} & e^{2(\alpha_1-j2\pi f_0)/f_r} & e^{2(\alpha_1+j2\pi f_1)/f_r} & e^{2(\alpha_1-j2\pi f_1)/f_r} & e^{2(\alpha_2+j2\pi f_2)/f_r} & e^{2(\alpha_2-j2\pi f_2)/f_r} \\ e^{3(\alpha_0+j2\pi f_0)/f_r} & e^{3(\alpha_0-j2\pi f_0)/f_r} & e^{3(\alpha_1+j2\pi f_1)/f_r} & e^{3(\alpha_1-j2\pi f_1)/f_r} & e^{3(\alpha_2+j2\pi f_2)/f_r} & e^{3(\alpha_2-j2\pi f_2)/f_r} \\ e^{4(\alpha_0+j2\pi f_0)/f_r} & e^{4(\alpha_0-j2\pi f_0)/f_r} & e^{4(\alpha_1+j2\pi f_1)/f_r} & e^{4(\alpha_1-j2\pi f_1)/f_r} & e^{4(\alpha_2+j2\pi f_2)/f_r} & e^{4(\alpha_2-j2\pi f_2)/f_r} \\ e^{5(\alpha_0+j2\pi f_0)/f_r} & e^{5(\alpha_0-j2\pi f_0)/f_r} & e^{5(\alpha_1+j2\pi f_1)/f_r} & e^{5(\alpha_1-j2\pi f_1)/f_r} & e^{5(\alpha_2+j2\pi f_2)/f_r} & e^{5(\alpha_2-j2\pi f_2)/f_r} \end{bmatrix}. \tag{17}$$

If **B** is known and has full rank, the fundamental, sub-SO, and super-SO components in the synchrophasor can be decoupled and separated by solving the linear equation system. Then, sub-SO and super-SO mode parameters can be identified. Therefore, the frequency and damping factor of each component is important. The frequency of each component is obtained in Section 3.1. However, the damping factors for each component in **B** are unknown. Therefore, an iterative idea is introduced to realize the mode parameter identification of the damping factor.

The initial iterative matrix is defined as $\mathbf{B_1}$, and the damping factor of each mode in $\mathbf{B_1}$ is set to 0. $\mathbf{B_1}$ is expressed as

$$B_1 = \frac{1}{2N_p}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ e^{j2\pi f_0/f_r} & e^{-j2\pi f_0/f_r} & e^{j2\pi f_1/f_r} & e^{-j2\pi f_1/f_r} & e^{j2\pi f_2/f_r} & e^{-j2\pi f_2/f_r} \\ e^{2(j2\pi f_0)/f_r} & e^{-2(j2\pi f_0)/f_r} & e^{2(j2\pi f_1)/f_r} & e^{-2(j2\pi f_1)/f_r} & e^{2(j2\pi f_2)/f_r} & e^{-2(j2\pi f_2)/f_r} \\ e^{3(j2\pi f_0)/f_r} & e^{-3(j2\pi f_0)/f_r} & e^{3(j2\pi f_1)/f_r} & e^{-3(j2\pi f_1)/f_r} & e^{3(j2\pi f_2)/f_r} & e^{-3(j2\pi f_2)/f_r} \\ e^{4(j2\pi f_0)/f_r} & e^{-4(j2\pi f_0)/f_r} & e^{4(j2\pi f_1)/f_r} & e^{-4(j2\pi f_1)/f_r} & e^{4(j2\pi f_2)/f_r} & e^{-4(j2\pi f_2)/f_r} \\ e^{5(j2\pi f_0)/f_r} & e^{-5(j2\pi f_0)/f_r} & e^{5(j2\pi f_1)/f_r} & e^{-5(j2\pi f_1)/f_r} & e^{5(j2\pi f_2)/f_r} & e^{-5(j2\pi f_2)/f_r} \end{bmatrix}. \tag{18}$$

The number of synchrophasor data within 1 *sec* is assumed to be N_r. In this case, the equation systems of number N_r-6 + 1 can be constructed according to Eq. (16) because of an equation system containing six synchrophasor data.

The solution of the equation system is defined as $\eta(m)$, whose absolute value $|\eta(m)|$ can be expressed as

$$|\eta(m)| = \begin{bmatrix} A_0(m)H_{01} \\ A_0(m)H_{02} \\ A_1(m)H_{11} \\ A_1(m)H_{12} \\ A_2(m)H_{21} \\ A_2(m)H_{22} \end{bmatrix}. \ m = 0,\ 1,\ 2,\ \ldots,\ Nr\text{-}5. \tag{19}$$

According to Eq. (19), the fundamental, sub-SO, and super-SO components in the synchrophasor are decoupled and separated. The mode parameter identification process is briefly described as follows. The logarithm of $|A_0(m)H_{01}|$, $|A_1(m)H_{11}|$, $|A_2(m)H_{21}|$ is taken. Then, linear fitting is performed to obtain the slope a and intercept b of the straight line. The damping factor is equal to a.

$$\ln(|A_0(m)H_{01}|) = am + b \tag{20}$$

An iterative method is introduced to improve the identification accuracy of the damping factor and the amplitude of sub-SO and super-SO. This process is briefly described as follows. First, the calculated damping factors of each component are substituted into **B** to obtain the second iterative matrix $\mathbf{B_2}$. Each component in the synchrophasor is again separated to obtain the mode parameters. This process is repeated until the damping factor error obtained before and after the iteration is less than one-thousandth. The damping factor obtained in the last iteration is selected as the identification result.

$|A_0(m)H_{01}|$ is taken as an example, and the acquisition process of each mode amplitude is briefly described. H_{01} is calculated by Eq. (5) using the identification results of frequency and damping factor. Then,

Table 1
Mode parameters.

	A	f (Hz)	α	θ
Mode 1	1	50	0	$\pi/3$
Mode 2	0.2	28	0.2	$\pi/6$
Mode 3	0.4	72	0.1	$\pi/8$

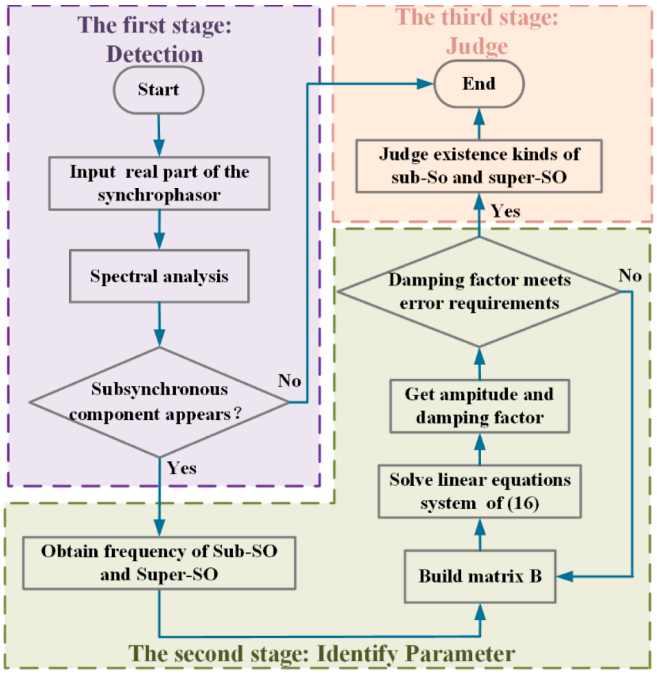


Fig. 3. Flowchart of the proposed method.

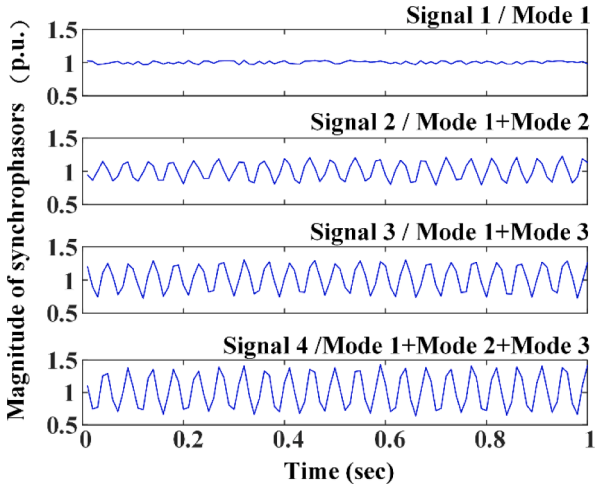


Fig. 4. Magnitude of synchrophasor.

A_0 is calculated using the intercept b obtained by the linear fitting in the last iteration, which is expressed as

$$A_0 = e^b / H_{01} \,. \tag{21}$$

3.3. Proposed sub-SO/super-SO detection and mode parameter identification method

The proposed method takes 2 secs as a data window. It utilizes the voltage or current synchrophasor to realize sub-SO and super-SO detection and mode parameter identification. Specific steps are listed as follows:

1) In the first half of the data window, the real part of the synchrophasor is selected as the feature quantity. Spectrum analysis is performed to detect the presence or absence of a sub-SO component.
2) If a sub-SO component exists, the time-domain oscillation has three cases. Subsequently, interpolated DFT is used to obtain the exact sub-SO component frequency. Then, the possible super-SO component frequency is calculated based on the frequency coupling.
3) The initial iterative matrix $\mathbf{B_1}$ is constructed from the obtained sub-SO and super-SO component frequency. The synchrophasor data of the second half data window are used to solve Eq. (16). The absolute value of the solution $|\eta(m)|$ is obtained. The oscillation amplitude and damping factor are calculated by using the method proposed in Section 3.2.
4) The amplitudes and damping factors of the sub-SO and super-SO components are obtained in the third step, which is used to judge the existence of sub-SO and super-SO in the time domain signal.

The proposed identification method in Section 3.2 is based on solving a linear equations system. Therefore, $\mathbf{B_1}$ is required to have full rank. $\mathbf{B_1}$ is a Vandermonde matrix; thus, it is full rank when the quantities in the second row are unequal. $\mathbf{B_1}$ is a singular matrix when the ratio of the fundamental frequency f_0 to the reporting rate f_r is a multiple of 0.5. Therefore, without changing the PMU device, a synchrophasor report method with a variable rate is required. The method has a data window of 2 secs, including the report rate $f_r = 2f_N$ Hz in the 1st *sec* and the report rate f_r less than $2f_N$ in the 2nd *sec*. In this study, the reporting rate is set to 1.6 f_N in the 2nd *sec*.

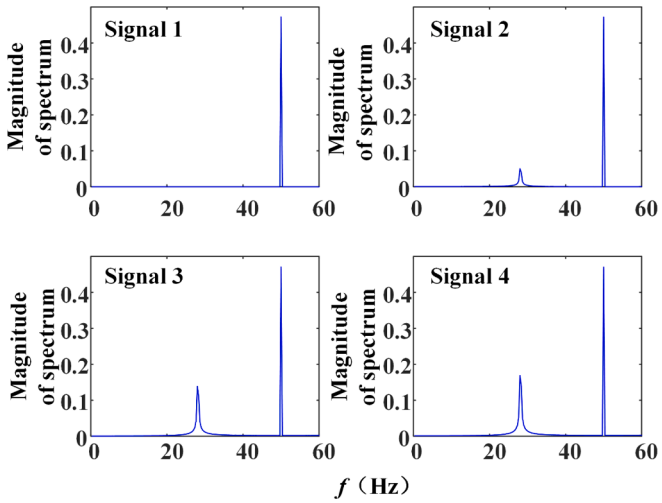


Fig. 5. Spectrum analysis of the real part of synchrophasor.

Table 2
Mode parameters.

	A	f (Hz)	α	θ
Mode 1	1	50	0	$\pi/9$
Mode 2	0.25	12	0.25	$\pi/4$
Mode 3	0.15	88	0.1	$\pi/8$
Mode 4	0.4	35	0.15	$\pi/6$
Mode 5	0.1	65	0.4	$\pi/3$

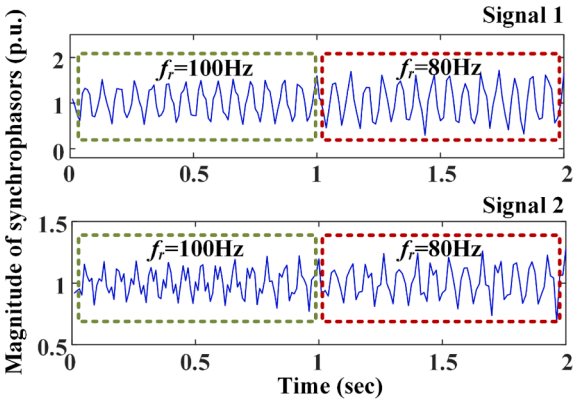


Fig. 6. Magnitude of synchrophasor.

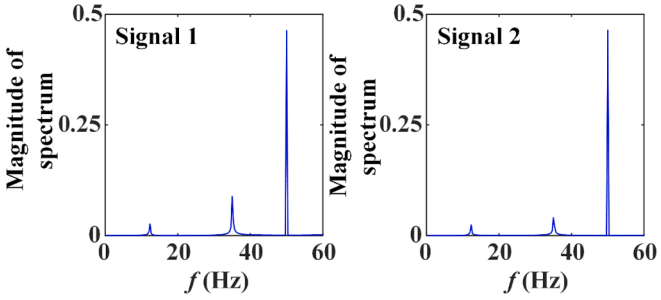


Fig. 7. Spectrum analysis of the real part of synchrophasor.

4. Method verification studies

4.1. Detection method verification

According to the above argument, the spectrum analysis is performed on the real part of the synchrophasor. If a sub-SO component is detected, three kinds of time-domain oscillation may exist. This section mainly verifies the validity of the detection method. The synthetic signal is modeled as Eq. (1), and its parameters are shown in Table 1. The synchrophasor corresponding to $x(t)$ is obtained by the DFT synchrophasor algorithm [21].

The three modes in Table 1 represent the fundamental, sub-SO, and super-SO components. The three modes are combined into four synthetic signals, as shown in Fig. 4. According to the latest research, the synchronous phasor contains non-Gaussian distribution noise [22]. Therefore, the 35 dB non-Gaussian distribution noise is added to the synchrophasor. The synchrophasor amplitudes of the four synthetic signals within 1 *sec* are shown in Fig. 4.

The spectrum analysis on the real part of the synchrophasor of the four synthesized signals is shown in Fig. 5. The spectral superposition of the sub-SO component with the spectral shift of the super-SO component in synchrophasor at $f_r = 2f_N$ is shown. The analysis in Section 3.1 is verified.

According to Fig. 5, the spectral analysis result of the real part of the synchrophasor can be utilized to judge whether a sub-SO and super-SO component exists in the synchrophasor. However, the existing kinds of sub-SO and super-SO components cannot be accurately judged. Meanwhile, the amplitude of each component obtained by the mode parameter identification method can be used to judge the existing kinds of sub-SO and super-SO components.

4.2. Verification of multiple pairs of frequency-coupled sub-SO/super-SO

The oscillation events may have the phenomenon in which the frequency-uncoupled sub-SO and super-SO or multiple pairs of the frequency-coupled sub-SO and super-SO exist in the voltage and current signal. The proposed method can be extended to this case. The mode parameter identification of this case is realized by increasing the number of equations in Eq. (16). For this case, the synthetic signal is modeled as

$$y(t) = \sum_{i=1}^{5} A_i e^{\alpha_i t} \cos(2\pi f_i t + \varphi_i) \tag{22}$$

Table 3
Identification results of mode parameters.

Mode		A	f / Hz	α
Mode 2	Theoretical results	0.25	12	0.25
	Identification results of signal 1	0.2521	12.0082	0.2501
	Identification results of signal 2	0.2521	12.0018	0.2499
Mode 3	Theoretical results	0.15	88	0.1
	Identification results of signal 1	0.1500	87.9918	0.0998
	Identification results of signal 2	0.0002	—	—
Mode 4	Theoretical results	0.4	35	0.15
	Identification results of signal 1	0.4026	34.9828	0.15
	Identification results of signal 2	0.0000	—	—
Mode 5	Theoretical results	0.1	65	0.4
	Identification results of signal 1	0.1021	65.0712	0.3987
	Identification results of signal 2	0.1013	64.9999	0.3995

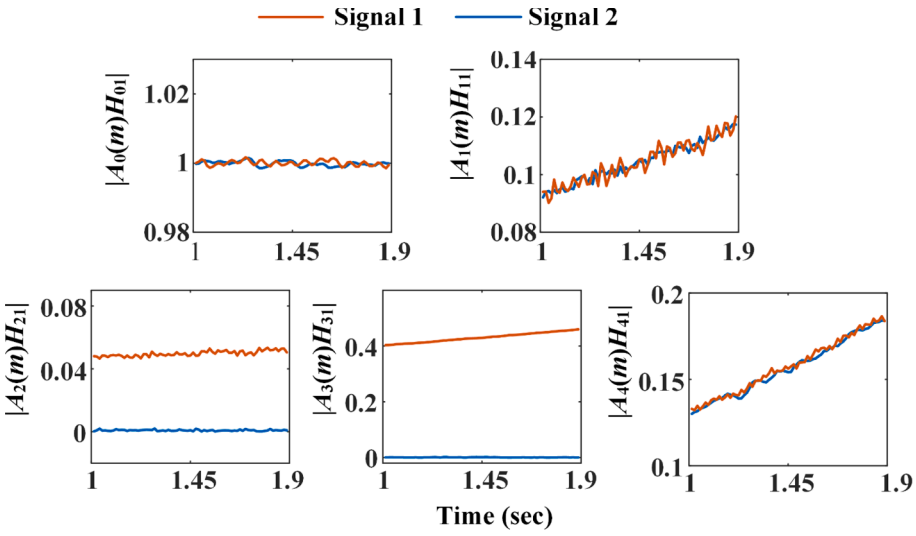


Fig. 8. The absolute value of the solution to a system of linear equations.

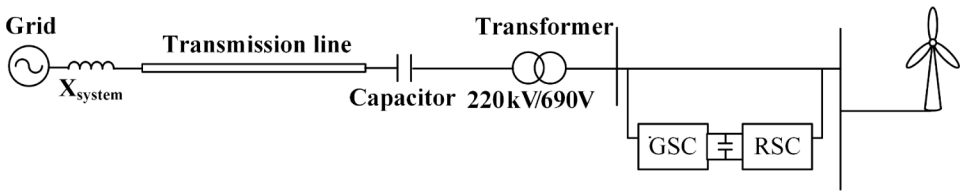


Fig. 9. Simulation model.

Table 4
Parameters of the model.

System Voltage	220 kV
Equivalent system reactance X_{system}	24.2 Ω
Transmission line inductance	0.78H
Transmission line resistance	9.68 Ω
Series capacitance (15 % compensation)	87.7 μF
Transformer voltages	220 kV / 690 V

Table 5
Simulated case conditions.

Case	Wind velocity	Series compensation percentage
1	10 m/s	40 %
2	10 m/s	55 %

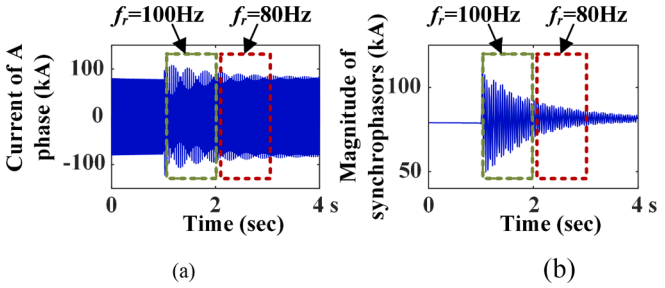


Fig. 10. Simulated results of case 1. (a) A-phase current wave and (b) the amplitude of synchrophasor.

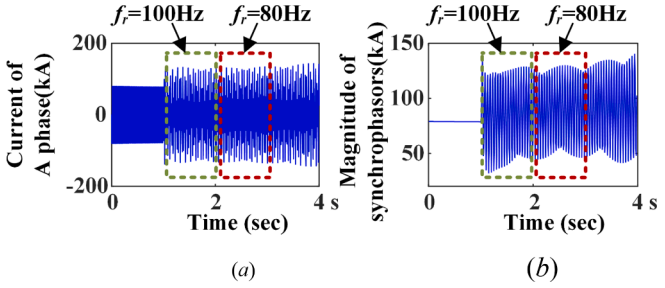


Fig. 11. Simulated results of case 2. (a) A-phase current wave and (b) the amplitude of synchrophasor.

where A_i, α_i, f_i, and φ_i represent the amplitude, damping factor, frequency, and initial phase of the ith mode component, respectively. Five modes exist in $y(t)$. Their parameters are defined in Table 2.

These five modes are combined into two synthetic signals. Signal 1 contains all modes in Table 2. Signal 2 contains the first, second, and fifth modes in Table 2. The synchrophasor amplitudes of the two signals within 2 secs are shown in Fig. 6.

Spectrum analysis is performed on the real part of the synchrophasor in the 1st *sec* of the two synthesized signals, as shown in Fig. 7. It proves

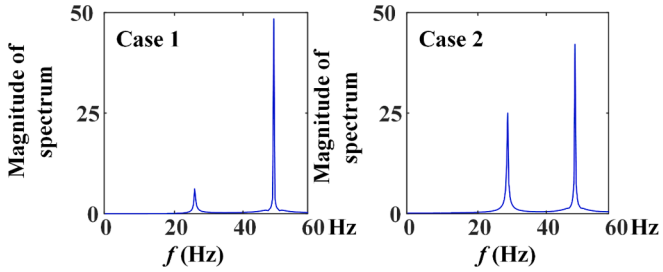


Fig. 12. Spectrum analysis of the real part of synchrophasor.

that the proposed method can be used in this case.

In this case, a tenth-order $\mathbf{B_1}$ is constructed to identify the sub-SO and super-SO mode parameters. Moreover, 80 synchrophasors in the 2nd *sec* are substituted into the equation system. The $|A_0(m)H_{01}|$, $|A_1(m)H_{11}|$, $|A_2(m)H_{21}|$, $|A_3(m)H_{31}|$, and $|A_2(m)H_{41}|$ of signals 1 and 2 are obtained via iteration. They are shown in Fig. 8.

As shown in Fig. 8, the third and fourth mode component amplitudes of signal 2 are 0. Signal 2 does not contain modes 3 and 4. Therefore, the proposed method can be applied to such complex situations. It also has good scalability. According to Fig. 3, the mode parameter identification results are shown in Table 3. These results verify that the identification results of the proposed method are accurate.

4.3. Verification with simulated data

The proposed method is further verified in a MATLAB/Simulink simulation model, as shown in Fig. 9. Here, the wind farm adopts doubly-fed wind turbines with a rated power of 100 MW. A filter is installed at the entrance of the grid-side converter (GSC) with a voltage level of 220 kV. The key parameters of the simulation model are shown in Table 4. The model's initial wind velocity is set to 10 m/s, and the initial series compensation of the line is 15 %.

The series compensation degree is increased at the 1st *sec*. Thus, the system occurs sub-SO and super-SO. The compensation degree settings of the two cases are shown in Table 5. The synchrophasor of 1–3 *sec* is taken as cases 1 and 2, respectively. The instantaneous data waveforms with a sampling frequency of 10 kHz are shown in Fig. 10(a) andFig. 11 11(a). The synchrophasor amplitudes reported to the WAMS main station are shown in Fig. 10(b) andFig. 11(b). The proposed method is applied to analyze the real part of the 1–2 *sec* synchrophasor in the two cases for sub-SO and super-SO detection. The analysis results are shown in Fig. 12.

According to Fig. 12, the presence of sub-SO and super-SO components in both simulation cases is illustrated. As shown in Fig. 12, the sub-SO and super-SO in case 2 are severe. Therefore, the proposed method is verified. It can detect sub-SO and super-SO. Iteration matrix **B** is constructed to obtain the mode parameters of sub-SO and super-SO. According to Fig. 3, the $|A_0(m)H_{01}|$, $|A_1(m)H_{11}|$, and $|A_2(m)H_{21}|$ of each component in both cases are obtained, as shown in Fig. 13.

According to Fig. 13, the proposed method can decouple and separate the components in the synchrophasor. As shown in Fig. 13, sub-SO and super-SO in case 1 show a decaying trend; by contrast, sub-SO and super-SO in case 2 show a divergent trend. Subsequently, the logarithms

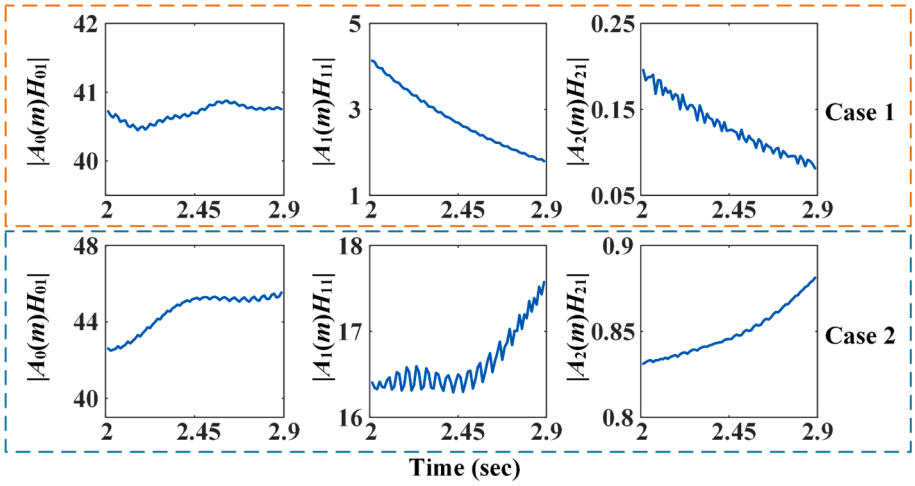


Fig. 13. The absolute value of the solution to a system of linear equations.

Table 6
Identification results of mode parameters.

Mode			*A* (kA)	*f* (Hz)	α
Case 1	Sub-SO	Reference results	32.2120	26.5578	0.9356
		Identification results of the proposed method	32.3430	26.5589	0.9242
		Identification results of [8]	33.1980	26.5581	0.9122
		Identification results of [17]	33.0976	26.5572	0.9433
	Super-SO	Reference results	1.4896	73.4422	0.8892
		Identification results of the proposed method	1.4926	73.4411	0.9095
		Identification results of [8]	—	—	—
		Identification results of [17]	—	—	—
Case 2	Sub-SO	Reference results	43.1280	29.751	0.0791
		Identification results of the proposed method	43.2380	29.756	0.0756
		Identification results of [8]	45.7206	29.754	0.0861
		Identification results of [17]	44.9870	29.756	0.0832
	Super-SO	Reference results	2.2312	70.2389	0.0702
		Identification results of the proposed method	2.1333	70.2440	0.0699
		Identification results of [8]	—	—	—
		Identification results of [17]	—	—	—

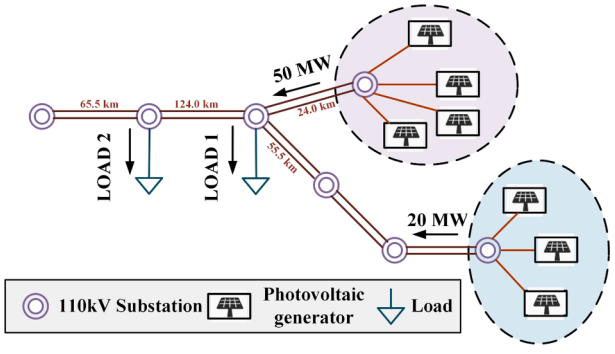


Fig. 14. Topology of the practical engineering case.

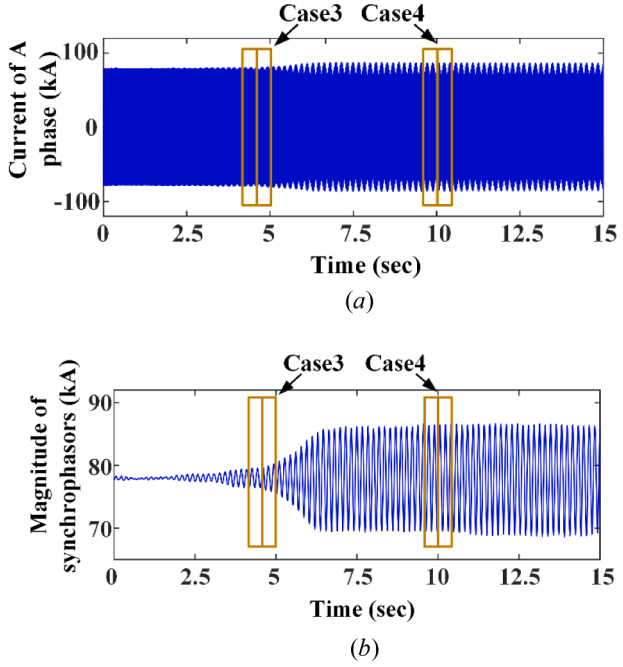


Fig. 15. Practical engineering case. (*a*) A-phase current real-time recording data and (*b*) A-phase current synchrophasor amplitude.

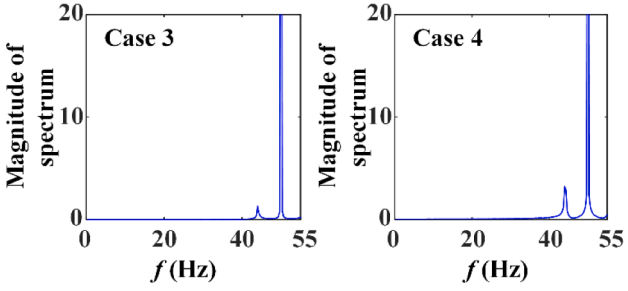


Fig. 16. Spectrum analysis of the real part of synchrophasor.

of $|A_0(m)H_{01}|$, $|A_1(m)H_{11}|$, and $|A_2(m)H_{21}|$ for cases 1 and 2 were taken to perform linear fitting and obtain the damping factor. Then, H_{i1} is calculated according to the identified frequency *f* and damping factor α. The amplitudes of each component are obtained. The sub-SO and super-SO identification results and comparison with the identification results of [8] and [17] are shown in Table 6. The reference values in Table 6 are obtained via the Hilbert–Huang transform (HHT) by processing the

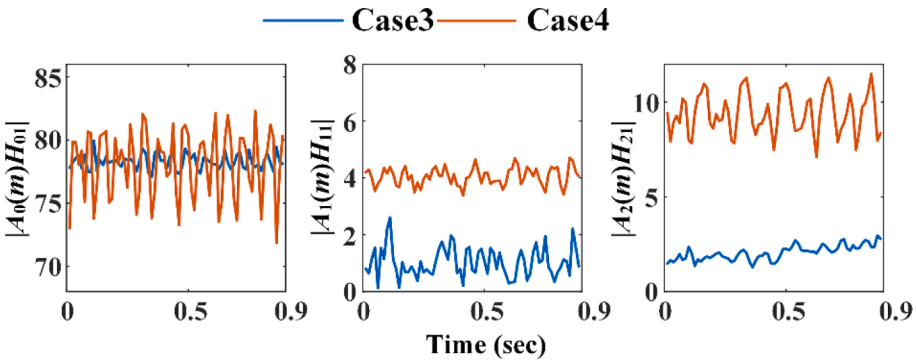


Fig. 17. The absolute value of the solution to a system of linear equations.

Table 7
Identification results of mode parameters.

Mode			A (kA)	f (Hz)	α
Case 3	Sub-SO	Reference results	0.8997	43.9971	0.0810
		Identification results of the proposed method	0.9054	43.9986	0.0764
	Super-SO	Reference results	0.8501	56.0101	0.7001
		Identification results of the proposed method	0.8496	56.0014	0.6860
Case 4	Sub-SO	Reference results	3.7120	44.1200	−0.0901
		Identification results of the proposed method	3.7141	44.1273	−0.0819
	Super-SO	Reference results	9.6001	55.8604	0.0000
		Identification results of the proposed method	9.5333	55.8727	−0.0001

selected fault recording data within 2–3 secs.

According to Table 6, the amplitude of the super-SO component is much smaller than that of the sub-SO component in both cases. The proposed method is compared with the methods in [8] and [17]. The identification results obtained in the present study are closer to the reference value. These results indicate that the existence of super-SO components prevents the Nyquist sampling theorem from being satisfied at $f_r = 100$ Hz, thereby leading to the spectral shift of the super-SO component and the spectral superposition of the sub-SO component. Therefore, the identification results of the methods in [8] and [17] have huge errors. The above finding verifies that the proposed method can distinguish the existing kinds of sub-SO and super-SO and accurately identify the mode parameters of sub-SO and super-SO components.

5. Verification with practical engineering case

The proposed method is verified by taking the sub-SO and super-SO fault recording data of a photovoltaic renewable energy station in the southwest area of China as an example. The photovoltaic renewable energy station has a rated active power output of 50 MW. Another photovoltaic station with a rated active power output of 20 MW is located nearby. The topology is shown in Fig. 14.

In the normal operation mode, the maximum power supply radius of the photovoltaic station is 330 km. The power supply area has two loads. The active power of loads 1 and 2 fluctuates in the ranges of 4–8 MW and 13–18 MW. During the oscillation process, the line voltage and line output active power fluctuate within 115.8–117.3 kV and 10–36 MW, respectively. Fig. 15 shows the real-time recorded data curve and the synchrophasor information of the A-phase current reported in the sub-SO and super-SO accidents, respectively.

In the practical engineering case, the frequency-coupled sub-SO and super-SO occur at 2.21 *sec*. The synchrophasors of 3–5 *sec* and 9–11 *sec* are taken as cases 3 and 4, respectively. The sub-SO and super-SO detection and the mode parameter identification are conducted by the proposed method.

The spectral analyses result of the real part of the synchrophasor for both cases are shown in Fig. 16. The PMU report rate f_r is 100 Hz, and the frequency resolution is 1 Hz. Thus, the precise frequencies of the two components identified by interpolated DFT are 43.9986 and 44.1273 Hz, respectively.

$\mathbf{B_1}$ is constructed using the identification result of the frequency according to Fig. 3. The 80 synchrophasors in the 2nd *sec* of the two cases are substituted into the equation system and solved. Then, the $|A_0(m)H_{01}|$, $|A_1(m)H_{11}|$, and $|A_2(m)H_{21}|$ of each component in both cases are obtained by iteration, as shown in Fig. 17.

The logarithms of $|A_0(m)H_{01}|$, $|A_1(m)H_{11}|$, and $|A_2(m)H_{21}|$ of cases 3 and 4 are taken, and linear fitting is performed to obtain the damping factor. Then, H_{i1} is calculated according to the identified frequency f and damping factor α. The amplitudes of each component are obtained. The sub-SO and super-SO identification results are shown in Table 7. The reference values in Table 7 are obtained by the HHT processing fault recording data within 1 *sec*.

Figs. 16 and 17 and Table 7 show that the proposed method can apply the synchrophasors to detect sub-SO and super-SO and identify the mode parameters accurately, further verifying the effectiveness of the proposed method.

6. Conclusion

Based on the synchrophasor data of the WAMS main station, the method for sub-SO and super-SO detection and mode parameter identification is proposed in this study to solve the sub-SO and super-SO problem caused by renewable energy generations and power electronic devices. The spectrum of the super-SO component is shifted, and the sub-SO component is superimposed when the reported rate $f_r = 2f_N$. The spectrum analysis of the real part of the synchrophasor can effectively detect whether the sub-SO and super-SO occurs. The proposed method can realize the detection and mode parameter identification of the sub-SO and super-SO components with uncoupled frequency or sub-SO and super-SO components with multiple frequency pairs. The proposed method is proved to exhibit good scalability.

CRediT authorship contribution statement

Yuhong Wang: Conceptualization, Methodology, Writing – review & editing. **Hongyu Wang:** Validation, Writing – original draft. **Yuyan Song:** Writing – review & editing. **Jianquan Liao:** Project administration, Supervision, Resources, Writing – review & editing. **Chunsheng Guo:** Writing – review & editing. **Zongsheng Zheng:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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